

Basic Quantitative Methods

Formula Sheet - Closed-book Written Exam Attachment

This sheet is cumulative. Each section identifies the lecture in which its material is first introduced. Definitions, conditions, units, and interpretation remain examinable even when a formula is provided.

Notation used throughout

- Ω : sample space; A, B : events; A^c : complement.
- X, Y : random variables; x_i, y_i : observed values for row i .
- n : sample size; N : population size.
- $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$, and p : population parameters.
- $\bar{X}, S^2, \hat{P}$: estimators; $\bar{x}, s^2, \hat{p}$: realised estimates.
- α : significance level; df : degrees of freedom.

Unless stated otherwise, tests on this sheet are **two-sided**.

Lecture 1 - Data summaries and units

For a category with count n_A among n observations,

$$\text{proportion} = \frac{n_A}{n}, \quad \text{percentage} = 100 \frac{n_A}{n} \%.$$

For values $x_1, \dots, x_n$,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \text{range} = \max_i x_i - \min_i x_i.$$

A change from $a\%$ to $b\%$ is $b - a$ **percentage points**. Its relative percentage change is

$$100 \frac{b - a}{a} \% \quad (a \neq 0).$$

Keep units throughout: a slope or rate has **outcome units per predictor unit**.

Lecture 2 - Probability foundations

$$0 \leq P(A) \leq 1, \quad P(\Omega) = 1, \quad P(A^c) = 1 - P(A).$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

For a finite sample space with equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

A random variable is a function $X : \Omega \rightarrow \mathbb{R}$. An event is a subset of Ω , not a numerical variable.

Lecture 3 - PMFs, CDFs, and expectation

For a discrete random variable with support $\mathcal{X}$,

$$p_X(x) = P(X = x), \quad p_X(x) \geq 0, \quad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

The cumulative distribution function (CDF) is

$$F_X(a) = P(X \leq a) = \sum_{x \leq a} p_X(x).$$

For any CDF,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

Expectation and transformed expectation are

$$E[X] = \sum_{x \in \mathcal{X}} x p_X(x), \quad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

If $X \sim \text{Bernoulli}(p)$,

$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad E[X] = p.$$

Lecture 4 - Variance and conditioning

Let $\mu = E[X]$. Then

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \{E[X]\}^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

For $X \sim \text{Bernoulli}(p)$,

$$\text{Var}(X) = p(1 - p).$$

The usual sample variance and SD are

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

For $P(B) > 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A | B)P(B).$$

If $B_1, \dots, B_k$ partition Ω ,

$$P(A) = \sum_{i=1}^k P(A | B_i)P(B_i).$$

Bayes' rule is

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_{j=1}^k P(A | B_j)P(B_j)}.$$

Lecture 5 - Joint PMFs and correlation

For discrete X, Y ,

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Marginal and conditional PMFs are

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Discrete X and Y are independent when, for every x, y ,

$$p_{X,Y}(x, y) = p_X(x)p_Y(y).$$

For independent events A, B with $P(B) > 0$, $P(A | B) = P(A)$.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}, \quad -1 \leq \rho_{XY} \leq 1.$$

Sample covariance and correlation are

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

Independence implies zero covariance when the relevant expectations exist; zero covariance does not generally imply independence.

Lecture 6 - Continuous distributions

For a continuous random variable with PDF f_X ,

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1, \quad P(X = x) = 0.$$

$$P(a < X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a).$$

If $X \sim U(a, b)$,

$$f_X(x) = \frac{1}{b-a} \quad (a \leq x \leq b), \quad P(c < X < d) = \frac{d-c}{b-a},$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Approximately 68% lies within $\mu \pm \sigma$, and 95% within $\mu \pm 1.96\sigma$.

Lecture 7 - Samples and estimators

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{P} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{for a 0/1 variable.}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad df = n-1.$$

The degrees of freedom follow from the constraint

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

For sampling without replacement, the course's independence approximation uses the 10% condition:

$$n \leq 0.10N.$$

Random sampling supports generalisation to the sampling frame; random assignment supports causal treatment comparisons.

Lecture 8 - Sampling distributions and SEs

For independent, identically distributed observations with mean μ and variance σ^2 ,

$$E[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{S}{\sqrt{n}}.$$

The CLT gives, for sufficiently large n ,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

For an independent normal sample with unknown σ ,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

For a sample proportion,

$$E[\hat{P}] = p, \quad \text{SE}(\hat{P}) = \sqrt{\frac{p(1-p)}{n}}.$$

Normal approximation working condition:

$$np \geq 10, \quad n(1-p) \geq 10.$$

Multiplying n by c^2 divides the SE by c .

Lecture 9 - Confidence intervals and zero tests

General confidence-interval form:

$$\text{estimate} \pm \text{critical value} \times \text{SE}.$$

For a population mean with unknown σ ,

$$\bar{X} \pm t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}}.$$

For 95% procedures, $z_{0.975} = 1.960$. Selected critical values are

df	5	10	20	30	60	∞
$t_{0.975, df}$	2.571	2.228	2.086	2.042	2.000	1.960

For a contrast $\delta = \mu - \mu_0$,

$$\hat{\delta} = \bar{X} - \mu_0, \quad T = \frac{\hat{\delta} - 0}{S/\sqrt{n}}, \quad df = n-1.$$

$$H_0 : \delta = 0 \quad \text{versus} \quad H_1 : \delta \neq 0.$$

The two-sided p-value is

$$p = P(|T_{df}| \geq |t_{obs}| \mid H_0).$$

Decision at level α : reject H_0 if $p \leq \alpha$; otherwise do not reject. For the same two-sided t procedure, the level- α test rejects zero exactly when the $100(1-\alpha)\%$ interval excludes zero.

Lecture 10 - Tests, errors, and power

For $H_0 : p = p_0$ against $H_1 : p \neq p_0$, define $\delta_p = p - p_0$. The null test statistic is

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \quad p\text{-value} = 2P(Z \geq |z_{obs}|).$$

Check $np_0 \geq 10$ and $n(1-p_0) \geq 10$ for the test. A large-sample proportion interval is

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The equivalent interval for $p - p_0$ subtracts p_0 from both endpoints.

Type I error: reject H_0 when H_0 is true,

$$P(\text{Type I error}) = \alpha,$$

Type II error: do not reject H_0 when a specified H_1 is true,

$$\text{Power} = P(\text{reject } H_0 \mid H_1 \text{ is true}).$$

For a regression coefficient row,

$$t = \frac{b_j - 0}{\text{SE}(b_j)}, \quad b_j \pm t_{1-\alpha/2, df} \text{SE}(b_j),$$

tests $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$. In R output: **Estimate** is b_j , **Std. Error** is $\text{SE}(b_j)$, **t value** is their ratio, and **Pr(>|t|)** is the two-sided p-value.

Always report the estimate and units, uncertainty, decision, population/model scope, practical importance, and causal limit.